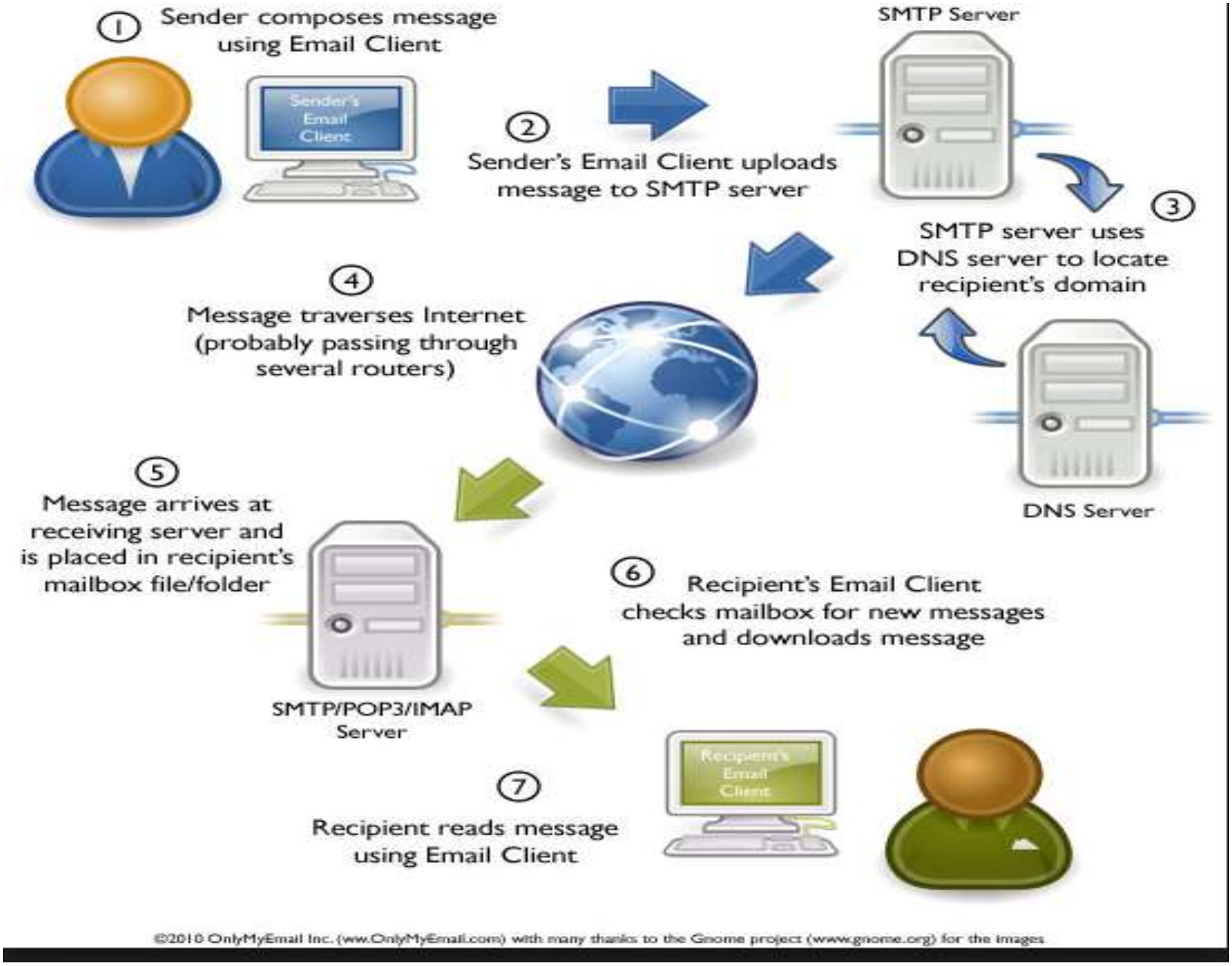
How e-Mail works
Demystifying your email server

Components of sending and receiving email

- Sender workstation
- SMTP Server
- Email messages storage
- Receiver workstation


1 Sender composes message using Email Client
Sender's Email Client
2 Sender's Email Client uploads message to SMTP server
SMTP Server
3 SMTP server uses DNS server to locate recipient's domain
DNS Server
4 Message traverses Internet (probably passing through several routers)
5 Message arrives at receiving server and is placed in recipient's mailbox file/folder
SMTP/POP3/IMAP Server
6 Recipient's Email Client checks mailbox for new messages and downloads message
Recipient's Email Client
7 Recipient reads message using Email Client
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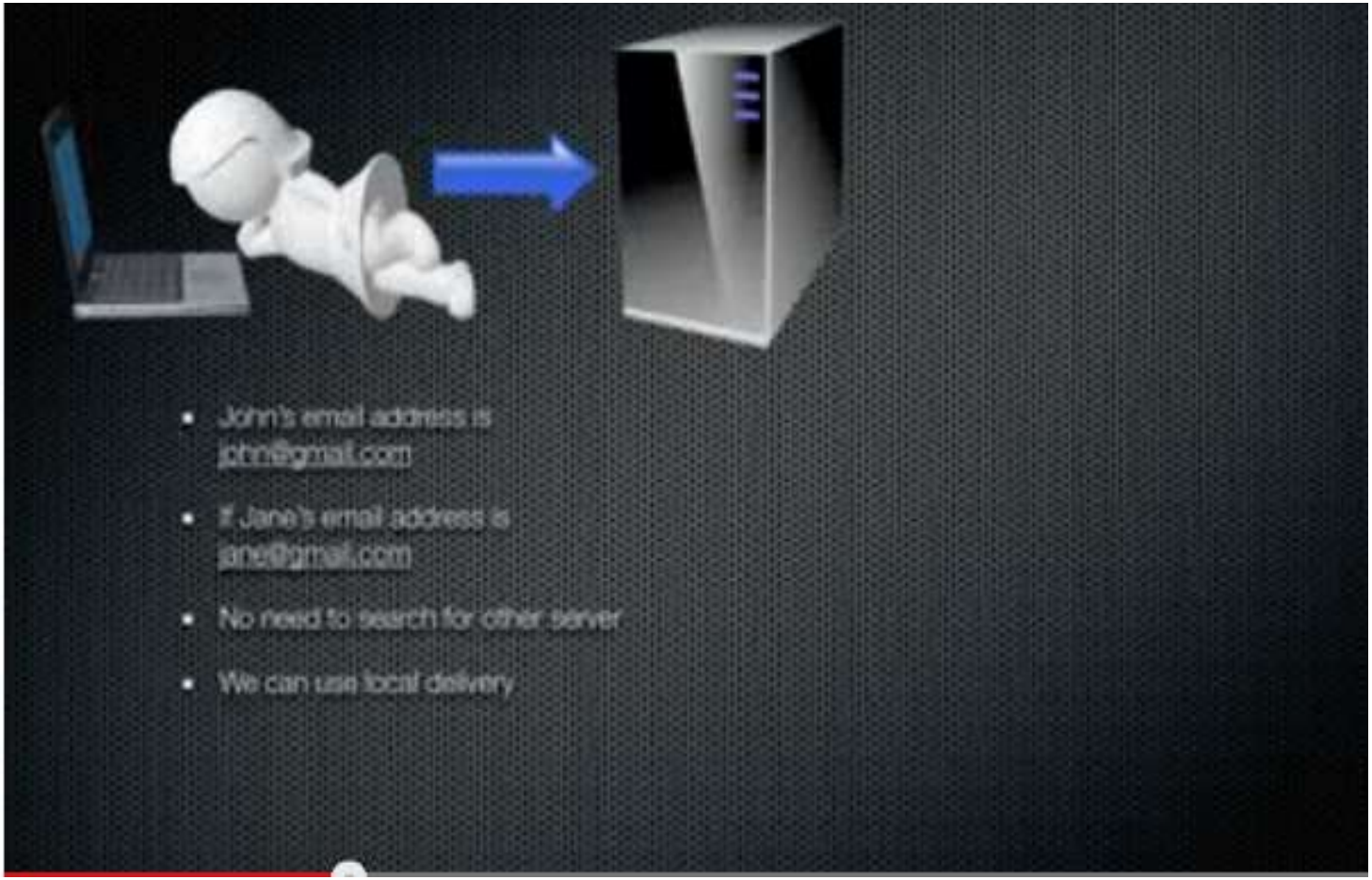
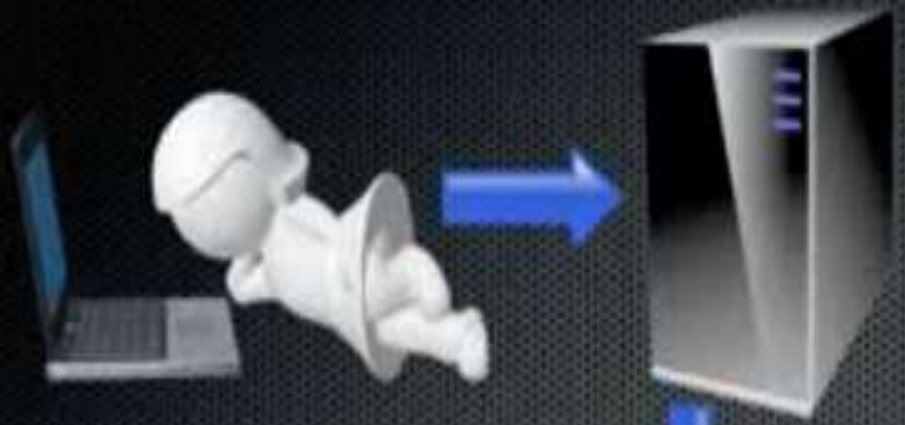
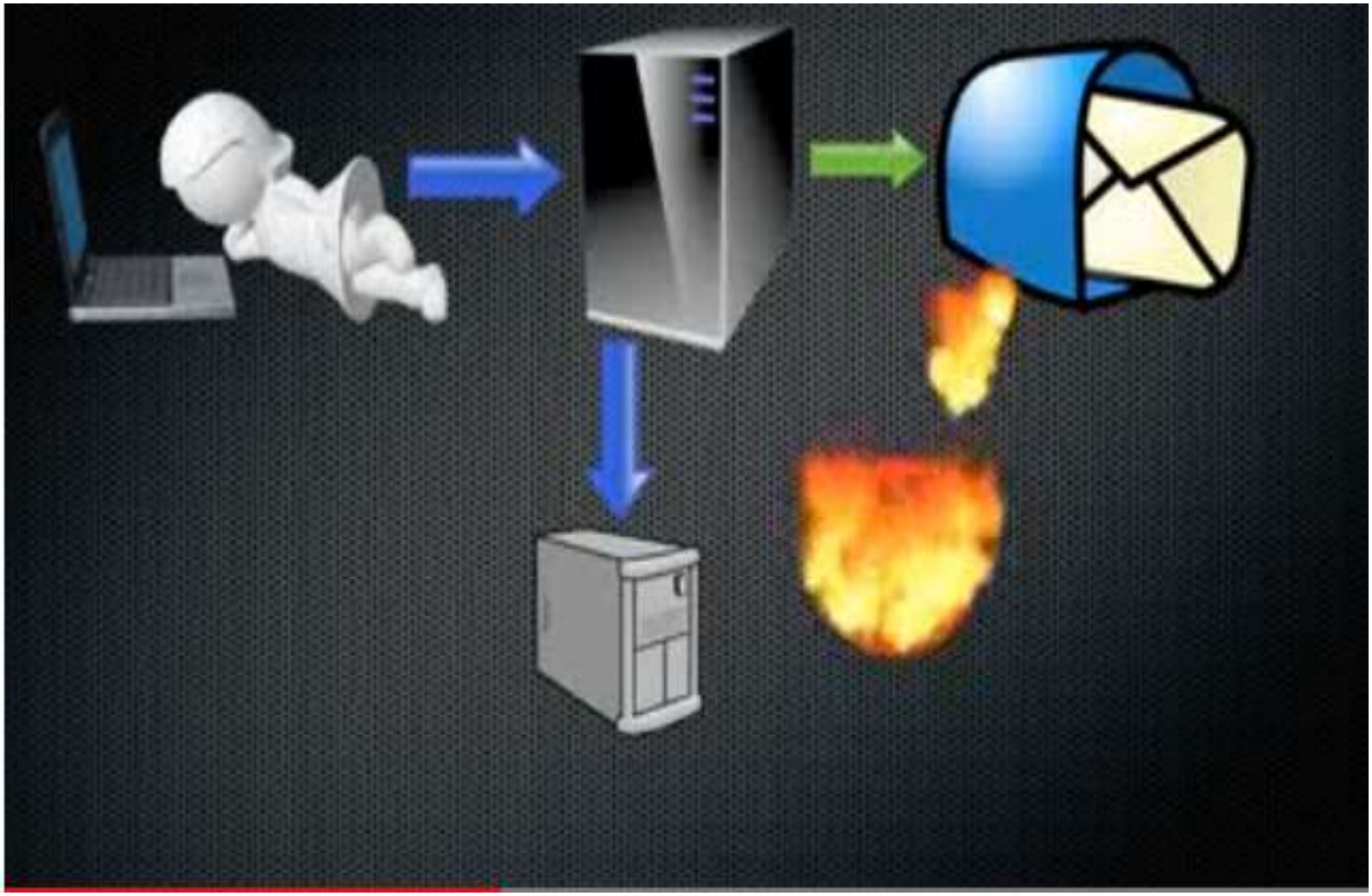
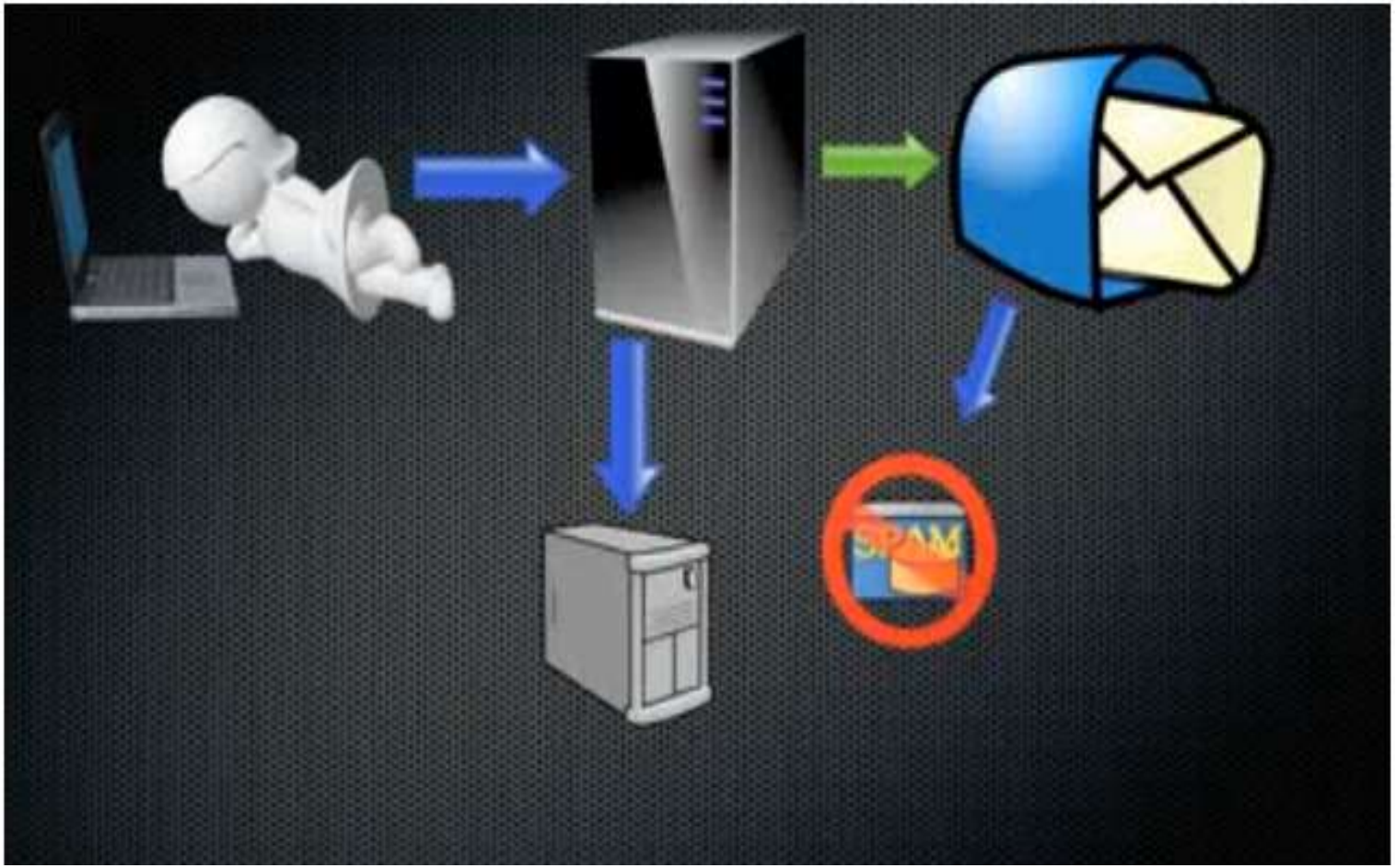

John's email address is
john@gmail.com
If Jane's email address is
jane@gmail.com
No need to search for other server
We can use local delivery



- DNS has a huge database of all host names and their IP addresses
- In this example John's email is: john@yahoo.com
- Take away john@
- Search for yahoo.com in host name database




SPAM

So what is SMTP? Why is it called SMTP? What is behind these letters?

- SMTP is an abbreviation for Simple Mail Transfer Protocol
- Maybe there is a Complex one somewhere?
- It was defined for the first time in 1982 by IETF

SMTP for sending email

- There was no Skype
- No instant messengers of any kind
- No email, just regular mail
- Delivered with a postman
- Yeah, depressing, I know

Welcome SMTP, hello 21st century

- The protocol is text only
- But it is very extendable
- So we can use it to send pictures, videos and other files
- By using extensions

The protocol is standard

- This protocol is a standard
- All email servers on the planet use it to send email
- Well, except maybe for Microsoft
- They have their own standards
- ...As usual

Let us see a real
example of using
SMTP

```
250-STARTTLS
250-CHUNKING
250-8BITMIME
250 BINARYMIME
MAIL FROM: <grvitaly@gmail.com>
250 2.1.0 grvitaly@gmail.com Address Okay
RCPT TO: <grvitaly@gmail.com>
250 2.1.5 grvitaly@gmail.com Address Okay
DATA
354 Start mail input; end with <CRLF>.<CRLF>
Subject: Test
Date: 10 Oct 2011

Hi,
his is a test message
.
250 Ok
```

Spam checks, very high level

- Check your IP against real time database
- Check your headers are correct
- Check your return address
- Check your message for stop words pattern